

What is a Number Sequence

Number sequences in Dynamics 365 F&O are essentially a series of unique identifiers assigned to records within the system. They distinguish and categorize various transactions, such as sales orders, purchase orders, invoices, and financial journals. Organizations can maintain accurate records and facilitate smooth data retrieval by assigning a unique sequence to each transaction.

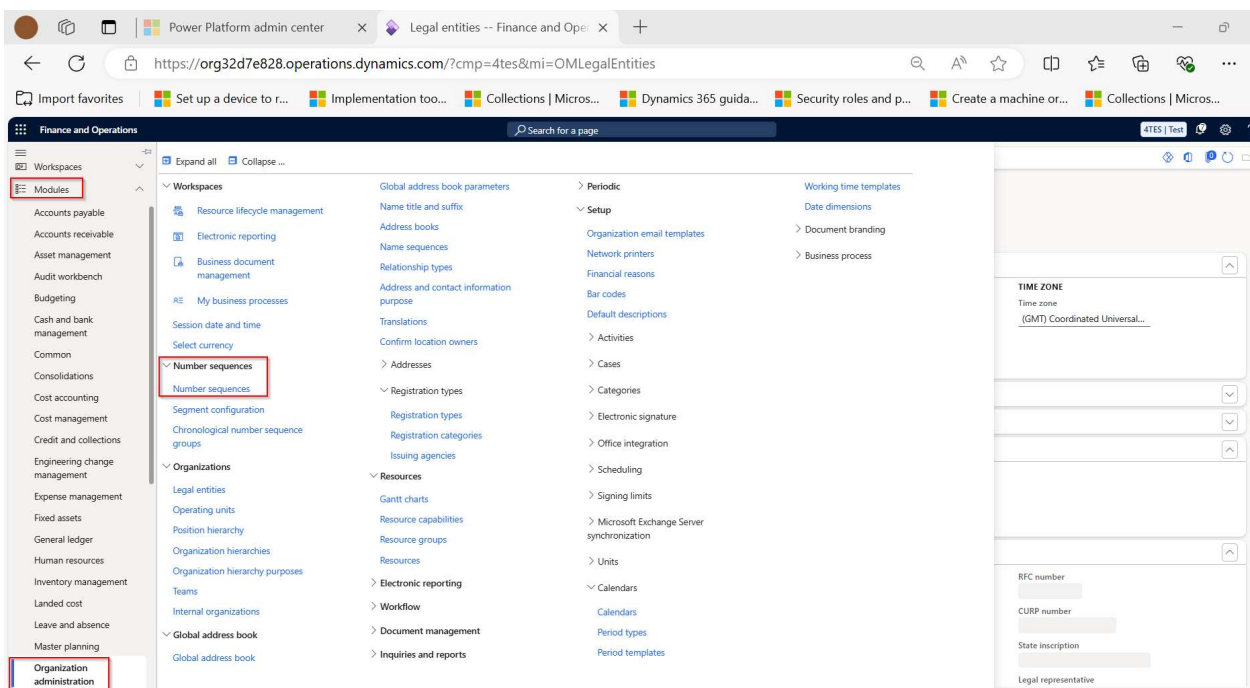
Number Sequences can be created in two different ways:

1. Using a Wizard
2. Manually

Using a Wizard

You would generally use this Wizard for a new legal entity or bulk creation of default number sequences.

- Go to Module > Organization administration > Number sequences > Number sequences.



Finance and Operations | Organization administration > Number sequences > Number sequences

← Number sequence Options 🔍

New
 Number sequence
Generate

Maintain
 Edit

Administration
 Status list
 Manual cleanup ▾
 History

Number sequences

Standard view ▾

☐	Number se...	Name	Smallest	Largest	Next	Format
☐	@Cus_1	@Cus_1	1	999999	11	#####
	<Tes_100	<Tes_100	1	99999	1	GLSI-####
	<Tes_101	<Tes_101	1	99999	1	INMF-####
	<Tes_102	<Tes_102	1	99999	1	ITCO-####
	<Tes_103	<Tes_103	1	99999	1	JPMF-####
	<Tes_104	<Tes_104	1	99999	1	MXMF-####
	<Tes_105	<Tes_105	1	99999	1	MYMF-####

This is a global setup (Shared Form), so the NS of every LE will be visible on the screen.

- Select **Generate** to initiate the Wizard.
- The Wizard is divided into **3 steps**, and all you need to do is review the process and click Next.

Standard view ▾

Set up number sequences

- ☒ Welcome
- ☐ Setup
- ☐ Completed

This wizard helps you create the required number sequences in your application setup.

The application determines which number sequences have not been specified and includes only these in the list.

You can choose to accept the application proposal for setup of each of the unspecified number sequences.

You can also choose to change names, values or other setup parameters, before the application creates your number sequences.

If you do not want the application to create a specific number sequence, you can just delete it in the list.

[Back](#)[Next](#)[Cancel](#)

- On this page, you can modify the identification code, the lowest value, and the highest value. In addition, you can indicate whether the number sequence must be continuous.

Standard view 

Set up number sequences

Welcome
 Setup
 Completed

Delete
 Include scope in format
 Remove scope from format

		OK	Area	Reference	Number sequ...	Smallest	Largest	Continuous	Format	
		☒	Accounts payable	Vendor account	Acco_28503	1	999999	☐	*****	
		☒	Accounts payable	Vendor account	Acco_28504	1	999999	☐	DLC-*****	
		☒	Accounts payable	One-time supplier	Acco_28505	1	999999	☐	ABC-*****	
		☒	Accounts payable	One-time supplier	Acco_28506	1	999999	☐	DLC-*****	
		☒	Accounts payable	1099 report number	Acco_28507	1	999999	☐	DLC-*****	
		☒	Accounts payable	Note ID	Acco_28508	1	999999	☐	ABC-*****	
		☒	Accounts payable	Note ID	Acco_28509	1	999999	☐	DLC-*****	
		☒	Accounts payable	Exchange rate adjust...	Acco_28510	130000000	139999999	☐	ABC-*****	
		☒	Accounts payable	Exchange rate adjust...	Acco_28511	130000000	139999999	☐	DLC-*****	
		☒	Accounts payable	Payment voucher	Acco_28512	140000000	149999999	☐	ABC-*****	
		☒	Accounts payable	Payment voucher	Acco_28513	140000000	149999999	☐	DLC-*****	
		☒	Accounts payable	Closing voucher	Acco_28514	150000000	159999999	☐	ABC-*****	
		☒	Accounts payable	Closing voucher	Acco_28515	150000000	159999999	☐	DLC-*****	

Back
 Next
 Cancel

UNIQUE KEY FOR ONE-TIME VENDORS. THE KEY IS USED WHEN CREATING NEW ONE-TIME VENDORS.

- You can make a number sequence Continuous by selecting the option. To add a scope segment to the format of a number sequence, choose the format from the list and then select Include Scope in format, or to remove it, select Remove Scope in the format.
- To exclude a number sequence from automatic generation, select the number sequence in the list and then select Delete.
- At the bottom of the table, you will find short descriptions of each and every number sequence.

- Once you click Next, you will be taken to the page where you can see the summary of all the new number sequences created. Click Finish to complete the process.

Standard view ▾

Set up number sequences

✓ Welcome

✓ Setup

● Completed

Area	Number of	⋮
Accounts payable	59	
Accounts receivable	47	
Advance holders	2	
Asset management	16	
Bank	24	
Basic	12	
Bill of materials	8	
Budget	10	
Commerce parameters	30	
Credit management	2	
Document management	2	
Fashion module	6	
Fixed asset	4	
Foreign trade	10	
General ledger	34	
Human resources	20	
India EXIM	9	
Internet	4	
Inventory management	48	

Back

Finish

Cancel

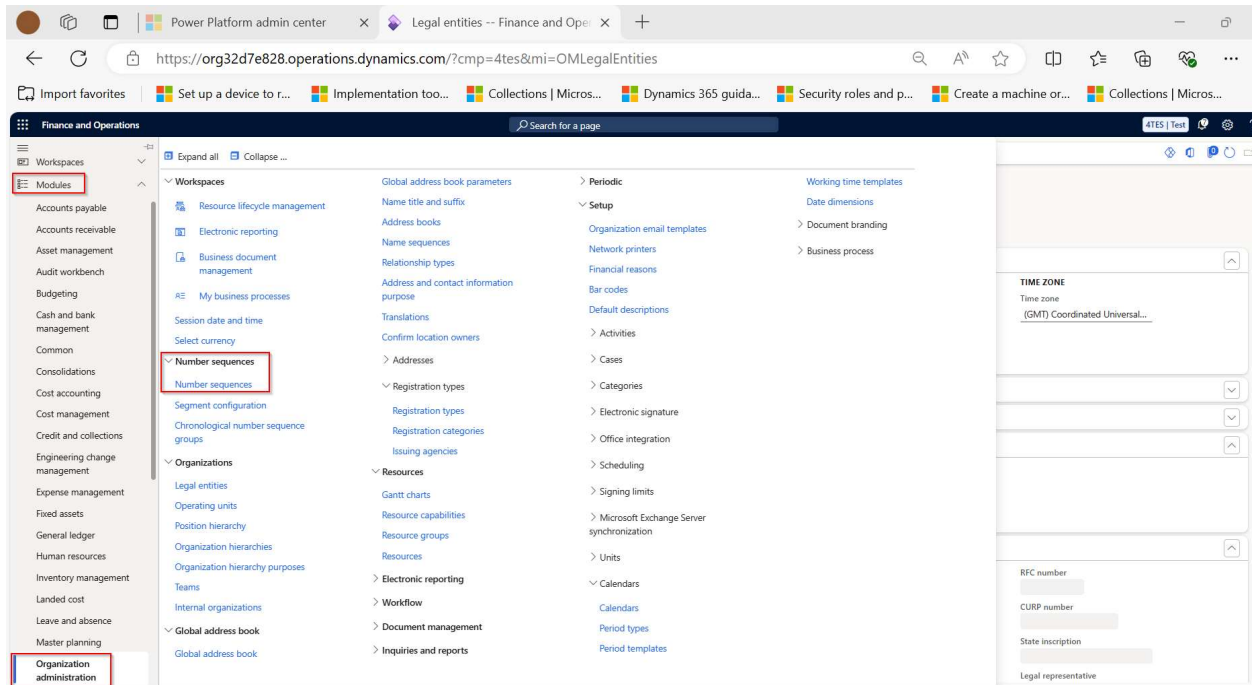
Note: This process can be used only once, i.e., when creating the Number. If you try to Generate number sequences using this option for the second time, the system will throw the error messages below.

Manual Process

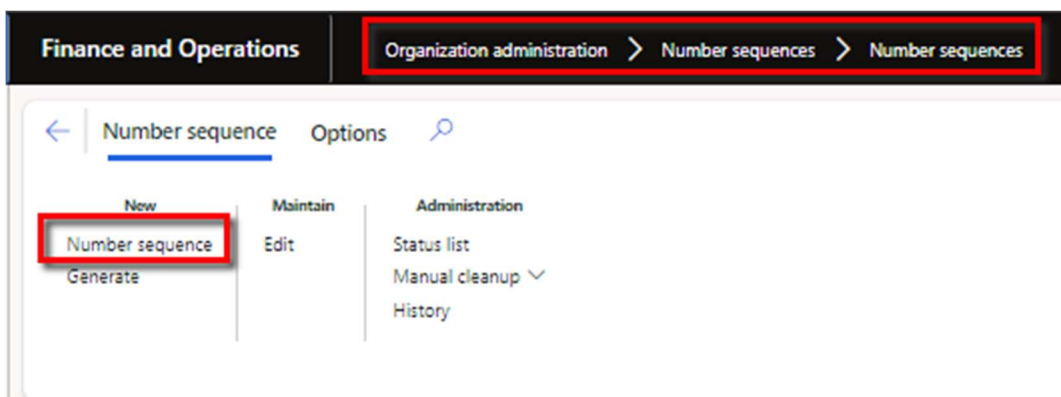
If you need to create a new number sequence manually, you can do it from two different places:

Option-1

- Go to Module > Organization administration > Number sequences > Number sequences.



Click on Number Sequence under the New option.



The screenshot shows the 'New Record' form for a Number Sequence in Microsoft Dynamics 365. The form is organized into several sections:

- Identification:** Contains two mandatory fields, 'Number sequence code' and 'Name', both highlighted with red rectangles.
- Scope parameters:** Contains a 'Scope' dropdown menu, also highlighted with a red rectangle.
- Segments:** A section for defining segments, with a table showing columns for Segment, Value, and Length.
- References:** A section for defining references, currently empty.
- General:** Contains various setup options, including 'In use', 'Manual', 'Continuous', 'Allow user changes', and 'Number allocation'.

- Fill in the Number Sequence Code and the Name under Identification FastTab. **(Mandatory)**
- Under the Scope Parameters FastTab, you will see multiple options to select from to decide the Scope of the number sequence. Following are the details of each option:
 - **Shared**—This option is used when a single number sequence is used for all organizations. The *shared Scope is available only for some reference/functions*.
 - **Company**—This option is used when a separate number sequence is used for each company. Use the Company scope when the underlying table includes the "DataAreaId" field, E.g., for the Customer Table. **(Select This option and select your company name)**
 - **Legal entity**—This option is used when a separate number sequence is to be used for each legal entity. Use the Legal entity scope when the underlying table does not include the DataAreaId field, e.g., for the Expense Report.
 - **Operating unit** - This option is used when a separate number sequence is used for each operating unit. Select the Operating unit after selecting this option. For example, select one of the departments or the cost centre.
 - **Company and fiscal calendar period**— *This option is used when a separate number sequence is used for each company and fiscal calendar period combination. A combination of the Company and Fiscal calendars needs to be selected here to proceed.*
 - **Legal entity and fiscal calendar period** - This option is used when a separate number sequence is used for each legal entity and fiscal calendar period combination.

- **Operating unit type** - A separate number sequence is used for each type of operating unit. This is a high-level number sequence for the Operating Unit option above. For example, select the whole department or the cost centre.
- Once you have selected the Scope, the next line will set the format of the number sequence under the **Segments** Fast Tab. So here, the fields will differ depending on the option selected in the Scope.

Number sequence code: VendCode Name: Vendor Code

Scope parameters

Scope: Company Company: 4TES

Segments

+ Add Remove ↑ Move up ↓ Move down

Segment	Value	Length
Company	4TES	4
Constant	-	1
Alphanumeric	#####	9

Format

4TES-#####

- In this section, we will set the format to add or remove the different segments, such as Company, Constant, or Alphanumeric. These options can be deferred according to the Scope selected.
- The following section is **Reference**, where we can either manually link the same or the system will populate this FastTab once we map the number sequence on the parameter form.

SETUP

In use: ☒ No Manual: ☒ No ALLOW USER CHANGES: To a lower number: ☒ No To a higher number: ☒ No

Stopped: ☒ No Continuous: ☒ No

NUMBER ALLOCATION

Smallest: 1 Largest: 999999 Next: 1

Performance

Preallocation - You should increase the preallocation quantity according to your regulatory requirement for better performance. The functionality preallocates numbers in NumberSequenceList for continuous number sequences and in memory of the computer for non-continuous number sequences. This applies to all the number sequences that are used as system keys, merely to link data in the database and sequences used for entities which cannot have gap in the numbers. You can use the Preallocation functionality for continuous and non-continuous number sequences. You must specify how many numbers to preallocate, e.g., 10 numbers. The system will select 10 numbers at a time to store in the memory of the computer. Thus, the system will select a new set of numbers from the database every tenth time a number is required. This functionality is very important for the general speed of the system. This is because selecting numbers from the database is considerably slower than selecting from the preallocated entity.

ACTIVATE PREALLOCATION

Preallocation: ☒ Yes Quantity of numbers: 5 Next: Largest:

- The next FastTab **General**, also called Setup, is the most essential part that determines the system's performance in high-volume transactions.
- There are multiple options/bullion to opt for; the **In Use Toggle** will be automatically set to **Yes** when it is mapped to a particular reference and generates one of the sequences.
- Use the **Stopped Toggle** if we do not want to use the number sequence anymore, and the system will not be able to generate numbers based on this number sequence.

- Use the **Manual Toggle** when you want the user to fill in the Number manually. If no format is set, it can be a defined number sequence. If the format is set, the user can still manually fill in the Number, but it has to follow the format.
- Under the **Allow User Changes section**, use the **To a lower number** Toggle when you want to allow users to edit the number sequence to a lower value than the already generated Number. Select the **To a higher number** Toggle to enable users to edit the number sequence to a higher value than the already generated Number.
- The **Number Allocation** option shows the **Smallest**, **Largest**, and **Next** numbers in the sequences. You can manually change the data in the Next field if you get an error about the already used voucher.
- Now, talking about the most critical Setup, **Continuous Toggle**, specify whether the number sequence is continuous or non-continuous.

A **continuous (CNS)** number sequence doesn't skip any numbers, but numbers may not be used sequentially. Numbers from a **noncontinuous (NonCNS)** number sequence are used sequentially, but the number sequence may skip numbers.

Deciding whether the number sequence should be a CNS or NonCNS is very important. This decision can, post-go-live, determine the performance of the system and the process.

You should select Continuous number sequence in case where there is legal compliance point or if it is a document to be used/shared out with stakeholders outside the organisation. For others it's advisable to use Non continuous, specially for internal documents/process where the data/process volume is high.

Note: From Version 10.0.36 Microsoft has brought a new feature which has resolved a long due issue of performance when the number sequence is in continuous. Feature is called 'Enable continuous number sequence performance improvements', it is not auto active.

If you will see the Performance fasttab was earlier only enabled for NonCNS number sequence only but now it is available also for CNS sequences just to ensure the performance matches that of a NonCNS.



- **Automatic Clean-up** FastTab appears only when you enable the **Continuous** option. This option, when enabled, cleans up the unused numbers at the interval defined in the Setup. By default, this is set at 24 hours, but we can move it to even so that the clean-up happens only on weekends. This way, the clean-up batch does not run during peak hours and does not happen in day-to-day operations.
- You can run the clean-up process manually or automatically to recover the lost numbers.

Automatic cleanup

Complete cleanup - periodic update

When you select the Continuous check box of a number sequence, the system automatically ensures that no numbers from the current sequence are lost. The system furthermore ensures that selected numbers are returned to the list if an update is unexpectedly terminated or fails.

In cases of power failure, crashes, or other kinds of unexpected termination, the system cannot return the numbers automatically. This is because the client process stops, leaving no time for the required cleanup of the selected numbers. The complete cleanup will find such temporary lost numbers and return them to the list with the status Free.

To execute the complete cleanup manually, click Clean up. You can also have the complete cleanup executed automatically with a specified time interval. To do this, select the Cleanup check box and specify an interval (in hours) for how often the cleanup should be performed.

ACTIVATE COMPLETE CLEANUP		LATEST COMPLETE CLEANUP	
Clean up	Interval	DateTime	
☒ Yes	24.00	2/20/2018 05:46:13 AM	

Performance

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This functionality is very important for the general speed of the system. This is because selecting numbers from the database is considerably slower than selecting from the preallocated entity.

ACTIVATE PREALLOCATION		ALLOCATED NUMBERS	
Preallocation	Quantity of number	Next	Largest
☒ Yes	5		

- **Performance FastTab** and **Preallocation** were earlier available only as a non-continuous option, but after the new update in version 10.0.36, this option is now available for continuous Sequence. This helps to block a set of numbers when a user uses the Sequence. For CNS cases, the blocked numbers are saved on the database, which was earlier not the case, leading to delays in processing and missing the numbers in between.

- Preallocation helps in case of a network issue where the AOS server connection is broken or in cases where multiple people are working on the same process. When you specify a quantity of numbers to preallocate, the system selects those numbers and stores them in memory.
- The **Quantity in Number** field can have a value based on the volume of data processed on a day-to-day basis; you need to analyse this well before configuring it on production. All the load testing can be done on the sandbox environment or other equivalent environment.

ADDITIONAL RESOURCES

<https://learn.microsoft.com/en-us/dynamics365/fin-ops-core/fin-ops/organization-administration/number-sequence-overview>

<https://learn.microsoft.com/en-us/shows/dynamics-365-fasttrack-architecture-insights/finance-and-operations-number-sequence>